

Chapter 5

Mathematical Induction

Introduction:

How do we know if something is true? For example, we have always studied that the sum of the internal angles in a triangle is 180° . But how do we know for sure? Or, say, we know $2 + 2 = 4$, always. No matter where, when, how we check it, it always turns out to be True. But can we prove that it is always going to be True? (In fact, it is quite difficult!) This is because, mathematics is quite different from other fields of science. If we conduct an experiment, say, 10 times and we get the same results, can we guarantee, we will always get the same results? What about 100 times? A million times? At what point do we know that the results will *always* be True? In mathematics, repeated experimentation is NOT a valid proof!

We would need to find a solid argument to show why the results will *always* be the same.

Let's look at some examples:

Claim 1: The sum of the first n positive integers: $(1 + 2 + 3 + 4 + \dots + n)$ is

$$\frac{n(n+1)}{2}$$

Let's check the results for a few simple cases:

$n = 1$:

Sum of 1st n numbers = 1

$$\frac{n(n+1)}{2} = 1$$

n = 2:

Sum of 1st n numbers = 1 + 2 = 3

$$\frac{n(n+1)}{2} = 3$$

n = 3:

Sum of 1st n numbers = 1 + 2 + 3 = 6

$$\frac{n(n+1)}{2} = 6$$

If the statement is true for the numbers we have tested, can we say it's true for all values of n?

Claim 2: If p is any prime number, $2^p - 1$ is also a prime

Let's check the results for a few sample numbers:

p = 2:

$$2^p - 1 = 3 \text{ (is prime)}$$

p = 3:

$$2^p - 1 = 7 \text{ (is prime)}$$

p = 5:

$$2^p - 1 = 31 \text{ (is prime)}$$

p = 7:

$$2^p - 1 = 137 \text{ (is prime)}$$

The statement seems to be true for all the numbers we have tested. Does that mean it is always true?

How about $p = 11$?

The question remains: How many cases do we need to check before we are sure that the claim is always true? No matter how many values of n we found the claim to be true for, there can always be a bigger value for which it is false. Hence, we need a general proof which covers all values of n .

Mathematical Induction is one way of doing this.

The trick used in Mathematical Induction is to consider the claim we are trying to prove as not a single claim, but a series of claims, one for each n .

We then prove the first statement in the sequence, and then prove that if any particular statement is true, then the one after it is also true. This enables us to conclude that all the statements are true.

Principle of Mathematical Induction

Induction or Principle of Mathematical Induction (PMI) is an extremely powerful proving technique for certain types of mathematical statements:

Statements that assert that something is true for all positive integers (or

*if n is T then
 $n+1$ is T
 n is T $\Rightarrow (n+1)$ is T*

natural numbers)/ all positive integers (or natural numbers) a particular point onwards.

Formally speaking, PMI is a technique used to prove that a mathematical statement, say, $P(n)$ holds for all positive integers/natural numbers n (or from a particular point onwards). To prove $P(n)$ is *always* true for all natural numbers using PMI, we first prove that $P(1)$ is true. We then assume that $P(k)$ is true for some natural number, k , and using this hypothesis, we prove that $P(k + 1)$ is also true. If $P(k + 1)$ is also True, then $P(n)$ becomes true for all natural numbers.

Mathematical Induction can be compared to the phenomenon of “**Falling Dominoes**”, where if one domino falls, it knocks down the second one, which knocks down the third one, and so on. In the end, all the dominoes are knocked down!

Principle of Mathematical Induction Statement

Any statement $P(n)$ which is for “ n ” natural number can be proved using the Principle of Mathematical Induction by following the below steps:

Step 1: Verify if the statement is true for trivial cases ($n = 1$) i.e. check if $P(1)$ is true.

Step 2: Assume that the statement is true for $n = k$ for some $k \geq 1$ i.e. $P(k)$ is true.

Step 3: If the truth of $P(k)$ implies the truth of $P(k + 1)$, then the statement $P(n)$ is true for all $n \geq 1$.

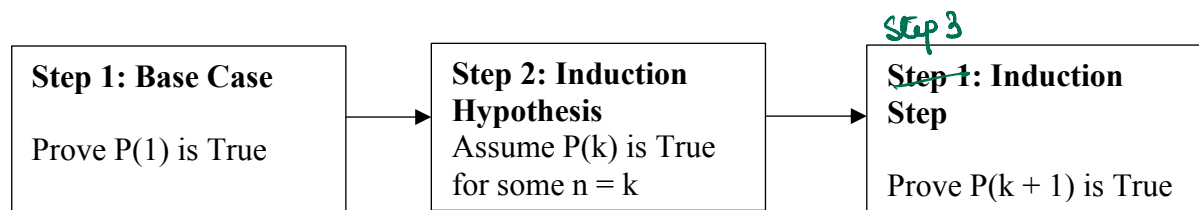


Fig 1

After proving these 3 steps, we can say that "By the Principle of Mathematical Induction, we have proved that $P(n)$ is true for all n ".

****Important point to remember about Step1:** Some mathematical statements hold true for n starting from a particular value, say, $n \geq 3$. In this case, to prove the result using the Principle of Mathematical Induction, step 1 will start from $n = 3$.

Let's revisit our claim1:

Prove the following claim using PMI:

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

$$\text{Let } P(n) \text{ be } 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$$

Base Step:

To prove $P(1)$ is true.

For $n = 1$, $LHS = 1$

$$RHS = 1(1+1)/2 = 2/2 = 1$$

Hence $LHS = RHS \Rightarrow P(1)$ is true.

Induction Hypothesis:

Assume that $P(n)$ holds for $n = k$, i.e., $P(k)$ is true

$$\Rightarrow 1 + 2 + 3 + 4 + 5 + \dots + k = \frac{k(k+1)}{2} \text{ --- (1)}$$

Induction Step:

Now we will prove that $P(k+1)$ is true.

$$\text{To prove: } 1 + 2 + 3 + 4 + \dots + (k+1) = \frac{(k+1)[(k+1)+1]}{2} = \frac{(k+1)(k+2)}{2}$$

$$\text{LHS} = 1 + 2 + 3 + 4 + \dots + (k+1)$$

$$= 1 + 2 + 3 + 4 + \dots + k + (k+1)$$

$$= (1 + 2 + 3 + 4 + \dots + k) + k+1$$

$$= \frac{k(k+1)}{2} + k+1 \text{ [Using (1)]}$$

$$= \frac{[k(k+1) + 2(k+1)]}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

$$= \text{RHS}$$

$$\Rightarrow P(n) \text{ is true for } n = k+1$$

Therefore, using PMI, we have proved that $P(n)$ is True for all natural numbers n .

**** If getting the RHS from the LHS seems difficult, simplify the LHS and the RHS separately and prove that they are equal.**

$$= \frac{k^2 + 2k + k + 2}{2}$$

$$= \frac{k^2 + 3k + 2}{2}$$

$$\frac{k^2 + k + 2k + 2}{2} = \frac{k^2 + 3k + 2}{2}$$

Let's look at another example:

Prove that $2^n > n$ for all positive integers n .

Assume $P(n)$: $2^n > n$

Base Step: To prove $P(1)$ is true.

For $n = 1$, we have $2^1 = 2 > 1$

$\Rightarrow P(1)$ is true.

$$\begin{aligned} 2^{k+1} &= 2^k \cdot 2^1 \\ &= 2^k \cdot 2 \\ &= 2 \cdot 2^k \end{aligned}$$

Induction Hypothesis: Assume that $P(n)$ holds for $n = k$, i.e., $P(k)$ is true

$\Rightarrow 2^k > k$ --- (1)

Induction Step: Now we will prove that $P(k+1)$ is true.

To prove: $2^{k+1} > k + 1$

$$2^{k+1} = 2 \cdot 2^k$$

$$2 \cdot 2^k > 2k \text{ [Using (1)]}$$

$$2k = k + k$$

$$k + k > k + 1 \text{ [Because any natural number other than 1 is greater than 1.]}$$

$$\begin{aligned} 2^{k+1} &= 2^k \cdot 2 = 2 \cdot 2^k \\ 2(2^k) &> 2k \\ 2k &= k+k \\ k+k &> k+1 \\ 2 \cdot 2^k &> (2k) > k+1 \\ 2 \cdot 2^k &> k+1 \\ 2^{k+1} &> k+1 \end{aligned}$$

$\Rightarrow P(n)$ is true for $n = k+1$

Therefore, by the Principle of Mathematical Induction, $P(n)$ is true for all natural numbers n .

Let's Practice:

Use mathematical induction to prove the following:

1. For all $n \geq 1$, prove that, $1^2 + 2^2 + 3^2 + \dots + n^2 = \{n(n+1)(2n+1)\} / 6$
2. $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = n(2n-1)(2n+1)/3$
3. Show that $10^{2n-1} + 1$ is divisible by 11 for all natural numbers.
4. For any positive integer n , prove that $n^3 + 2n$ is always divisible by 3

1. Let $P(n)$ be $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

Base Case:

For $n=1$

$$\text{LHS} = 1$$

$$\text{RHS} = \frac{1(1+1)(2 \times 1 + 1)}{6} = 1$$

$$\therefore \text{LHS} = \text{RHS}$$

$\therefore P(1)$ is true

Induction hypothesis

We assume that $P(n)$ is true for some $n=k$

$$\therefore 1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6} \dots (1)$$

Induction Step

To prove: $1^2 + 2^2 + 3^2 + \dots + (k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}$

LHS :

$$1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2$$

Using (1) $\rightarrow \frac{k(k+1)(2k+1)}{6} + (k+1)^2$

$$= \frac{k(k+1)(2k+1) + 6(k+1)^2}{6}$$

$$= \frac{(k+1)(k(2k+1) + 6(k+1))}{6}$$

$$= \frac{(k+1)(k(2k+1) + 6(k+1))}{6}$$

$$= \frac{(k+1)(2k^2 + \underbrace{k + 6k}_{7k} + 6)}{6}$$

$$= \frac{(k+1)(2k^2 + 4k + 3k + 6)}{6}$$

$$= \frac{(k+1) \left[2k \left(\frac{6}{k+2} \right) + 3(k+2) \right]}{6}$$

$$= \frac{(k+1)(2k+3)(k+2)}{6} = \text{RHS}$$

$\therefore \text{LHS} = \text{RHS}$

$\therefore P(n)$ is true for $n = k+1$

$\therefore$ By PMI $P(n)$ is true for all natural #s.

$$2. 1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{n(2n-1)(2n+1)}{3}$$

Let $P(n)$ be $1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{n(2n-1)(2n+1)}{3}$

Base Case

When $n=1$

$$\text{LHS} = 1^2 = 1$$

$$\text{RHS} = \frac{1(2 \times 1 - 1)(2 \times 1 + 1)}{3} = \frac{3}{3} = 1$$

$\therefore P(1)$ is True

Induction hypothesis

We assume that $P(n)$ is true for some $n=k$

$\therefore P(k)$ is true

$$\therefore 1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 = \frac{k(2k-1)(2k+1)}{3} \dots (1)$$

Induction step

To prove:

$$1^2 + 3^2 + 5^2 + \dots + (2(k+1)-1)^2 = \frac{(k+1)(2(k+1)-1)(2(k+1)+1)}{3}$$

$$\text{LHS: } 1^2 + 3^2 + 5^2 + \dots + (2k-1)^2 + (2(k+1)-1)^2$$

using (1)

$$\frac{k(2k-1)(2k+1)}{3} + (2(k+1)-1)^2$$

$$= \frac{k(2k-1)(2k+1)}{3} + (2(k+1)-1)^2$$

$$= \frac{(2k^2 - k)(2k+1) + 3(2k+1)^2}{3}$$

$$= \frac{(2k+1)(2k^2 - k + 3(2k+1))}{3}$$

$$= \frac{(2k+1)(2k^2 - k + 6k + 3)}{3}$$

$$= \frac{(2k+1)(2k^2 + 5k + 3)}{3}$$

$$= \frac{(2k+1)(2k^2 + 3k + 2k + 3)}{3}$$

$$= \frac{(2k+1)(k(2k+3) + 1(2k+3))}{3}$$

$$= \frac{(2k+1)(k+1)(2k+3)}{3} = \text{RHS}$$

$\therefore \text{LHS} = \text{RHS}$
 $\therefore P(k+1)$ is true $\therefore P(n)$ is true for $k+1$

$\therefore$ By PMI $P(n)$ is true for all natural #s

3. $10^{2n-1} + 1$ is divisible by 11 for all natural #s

Let $P(n)$ be $10^{2n-1} + 1 = 11m$

Base case

For $n=1$

$$\text{LHS} \rightarrow 10^1 + 1 = 11 = 11 \times 1$$

$\therefore P(1)$ is true

Induction hypothesis

Let $P(n)$ be true for some $n=k$

$$\therefore 10^{2k-1} + 1 = 11m \quad \dots (1)$$

Induction step

To prove : $10^{2(k+1)-1} + 1 = 11m'$

LHS :

$$10^{2k+2-1} + 1 = 10^2 \cdot 10^{2k-1} + 1 \rightarrow 10^2 (10^{2k-1}) + 1$$

$$= 10^2 \left(\underline{10^{2k-1} + 1} - 1 \right) + 1$$

$$= 10^2 (11m - 1) + 1 \quad \dots \text{using (1)}$$

$$= 10^2 \cdot 11m - 100 + 1$$

$$= 11 \cdot 10^2 m - 99$$

$$= 11(10^2 m - 9) = 11m'$$

$\therefore P(n)$ is true for $n=k+1$

$\therefore$ By PMI $P(n)$ is true for all natural #s.

4. $n^3 + 2n$ is always divisible by 3

$$\text{Let } P(n) \text{ be } n^3 + 2n = 3m$$

Base Case

When $n=1$

$$\text{LHS} = 1^3 + 2 \times 1 = 3 = 3 \times 1 = 3 \times m$$

$\therefore P(1)$ is true

Induction hypothesis

Let $P(n)$ be true for some $n=k$

$$\therefore k^3 + 2k = 3m \quad \dots (1)$$

Induction Step

To prove $P(k+1)$ is true

$$\text{i.e. } (k+1)^3 + 2(k+1) = 3m'$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$\text{LHS} : k^3 + 3k^2 \cdot 1 + 3k \cdot 1^2 + 1^3 + 2k + 2$$

$$= k^3 + 3k^2 + 3k + 2k + 3$$

$$= \underline{k^3 + 2k} + 3k^2 + 3k + 3$$

$$= 3m + 3(k^2 + k + 1) \quad \dots \text{Using (1)}$$

$$= 3(m + k^2 + k + 1) = 3m'$$

$\therefore P(n)$ is true for $n=k+1$

$\therefore$ By PMI $P(n)$ is true for all natural #s.